

1. Introduction

Image enhancement improves the visual appearance of an image in a subjective, perception-oriented manner, distinct from restoration, which aims to recover an image's original condition based on a degradation model. No single enhancement method is universally applicable; dark, bright, low-contrast, and blurred images each exhibit different characteristics, so applying the same method across all images can actually degrade quality. This assignment requires implementing enhancement operators from scratch, automatically classifying image types, and selecting appropriate methods based on that diagnosis.

2. Theoretical Basis

Point operations (negative, logarithmic, gamma, contrast stretching, histogram equalization, thresholding) depend only on the pixel's own value, while spatial operations (mean filter, Gaussian, unsharp masking, Laplacian) involve neighboring pixels through convolution. Result effectiveness is evaluated using four metrics: mean (brightness), standard deviation (contrast), Shannon entropy (information content), and Laplacian variance (edge sharpness).

3. Methodology

Fourteen 512×512 pixel images were automatically classified based on three statistical features through a sequential if-elif chain: dark (mean<85), bright (mean>170), low contrast (std<45), blurred (Laplacian variance<100), and normal for the remainder. Each label triggers a different processing branch.

4. Results and Analysis

Classification yielded 4 bright, 4 dark, 4 normal, and 2 low-contrast images not a single image was classified as blurred. For dark images, both logarithmic and gamma transforms increased brightness and generally entropy, though in one case an overly aggressive logarithmic transform actually reduced contrast and entropy. For bright images, the default gamma (0.4 which brightens) that should instead have darkened them caused severe saturation, reaching 43.9% saturated pixels in one image, while systematically reducing contrast and entropy. For low-contrast images, histogram equalization gave the most consistent contrast improvement, whereas thresholding destroyed information entirely (entropy collapsing near zero). The four images labeled normal received no processing at all, even though some were visibly blurred.

5. Discussion and Findings

The blur threshold is too loose. Calibration via progressive Gaussian blurring showed Laplacian variance dropping drastically (from thousands to tens) with just sigma 1.0–1.5,

so the threshold of 100 failed to catch images that were visibly blurred; a more appropriate threshold would be around 250–300.

The if-elif ordering causes misprioritization. The image with the lowest Laplacian variance in the entire dataset (the most blurred) fell into the low-contrast branch because the std condition was checked first, so the root cause (blur) was never addressed.

Histogram equalization damages color. Equalizing a combined histogram across the three RGB channels caused significant color shift (inter-channel differences increased 2–3x), producing rainbow noise this should instead be applied only to the luminance channel (e.g., YCrCb).

Other implementation flaws: the Laplacian sharpening function was never called despite being needed for blurred images; the "noise" classification branch can never be triggered; negative and thresholding operations in certain branches added no information whatsoever (evidenced by stagnant or collapsed entropy).

6. Conclusion

All operators were successfully implemented from scratch. Automatic classification worked effectively for dark/bright/low-contrast/normal but failed entirely for blurred images due to a loose threshold. No universal method exists — suitability depends on image type, and misdiagnosis (misclassification) directly leads to inappropriate method selection. Enhancement success must be judged using multiple metrics, since an increase in brightness can be accompanied by a decrease in contrast that makes the image look worse rather than better.

7. Recommendations

Raise the Laplacian variance threshold to 250–300; prioritize the blur check or use a weighted composite score; apply adaptive gamma (>1 for bright images, <1 for dark images); perform histogram equalization only on the luminance channel; activate the Laplacian sharpening function for blurred images; remove unused classification branches; and cap the maximum value in the logarithmic transform to prevent excessive saturation.